



KaxaNuk
Sharing knowledge

Strategy Presentation

Liquidity-Weighted Trend Strategy

Content



- Strategy Definition
- Feature Engineering
- Strategy Design

Strategy Definition

Liquidity-Weighted Trend Strategy

Idea Description



This strategy explores trend persistence in U.S. equities using a simple but robust signal.

The core idea is that stocks experiencing sustained upward price trends tend to continue outperforming due to:

- Gradual information diffusion
- Institutional trading flows
- Behavioral underreaction

Trend persistence is identified through a moving average crossover signal.

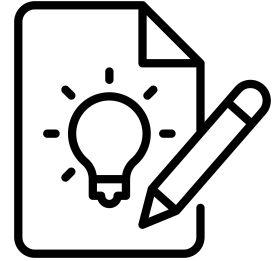
Stocks are considered bullish when:

- **SMA(50) > SMA(200)**

Eligible securities are then ranked by 3-month traded value, and the top 35 or X most liquid stocks are selected.

If fewer than 35 or X stocks satisfy the trend condition, remaining capital is allocated to:

- **SPY ETF** - SPDR S&P 500 ETF Trust



Background Research



Momentum and trend-following strategies have been extensively documented in financial research.

Key findings from academic literature:

Price Momentum

[Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency](#)

by Narasimhan Jegadeesh and Sheridan Titman (1993)

- Stocks that performed well in the past tend to continue outperforming in the short to medium term.
- One of the most widely replicated anomalies in finance.

Background Research



Trend Following Across Asset Classes

[Time Series Momentum](#) by Tobias Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen (2012)

- Trend-following strategies generate positive returns across equities, commodities, currencies, and bonds.
- Momentum effects persist across global markets.

Behavioral Explanation

[Investor Psychology and Security Market Under- and Overreactions](#) by Kent Daniel, David Hirshleifer, and Avanidhar Subrahmanyam (1998)

- Investor underreaction to new information can lead to persistent price trends.

Liquidity and Implementation

[Trading Costs of Asset Pricing Anomalies](#) by Andrea Frazzini, Ronen Israel, and Tobias J. Moskowitz (2014)

- Liquidity constraints and transaction costs are critical when implementing anomaly-based strategies.

Investable Universe and Constraints

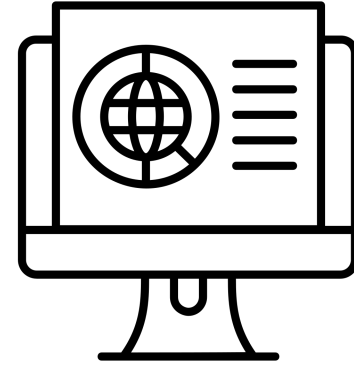


Universe:

→ KN US Equity Benchmark Historical Holdings

Strategy constraints:

- US listed equities
- Minimum trading history required
- Liquidity filter based on traded value
- Long-only portfolio implementation
- Maximum position size limits



These constraints ensure the strategy is realistic and investable.

Benchmark



Benchmark:

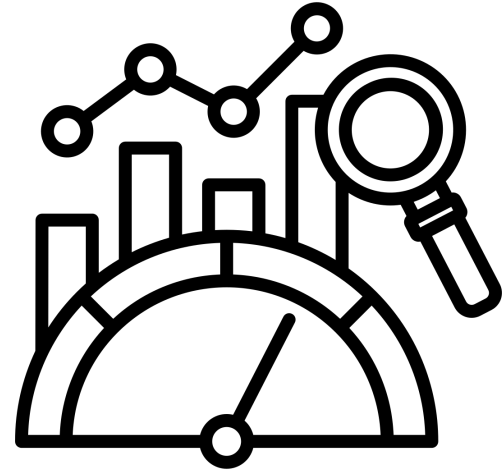
→ KN US Equity Benchmark

This benchmark provides:

- A transparent investable universe
- Consistent portfolio reference
- Historical benchmark returns

Benchmark comparison allows us to evaluate:

- Excess returns
- Tracking error
- Active performance



Hypothesis



The strategy is motivated by behavioral and structural market inefficiencies.

Behavioral drivers:

- Investors underreact to new information
- Investors chase past performance

Structural drivers:

- Institutional portfolio adjustments occur gradually
- Liquidity concentration reinforces momentum

Hypothesis:

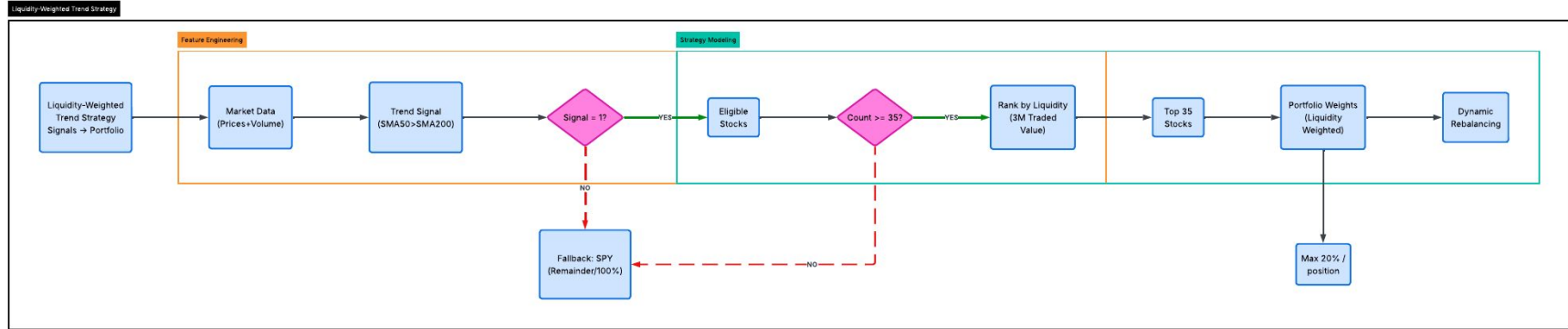
Stocks exhibiting sustained positive trends and strong liquidity will outperform the benchmark over intermediate horizons.

Null hypothesis:

The strategy does not outperform the benchmark.



Unified Modeling Language Diagrams



Feature Engineering

Liquidity-Weighted Trend Strategy

Data



The strategy relies primarily on market data.

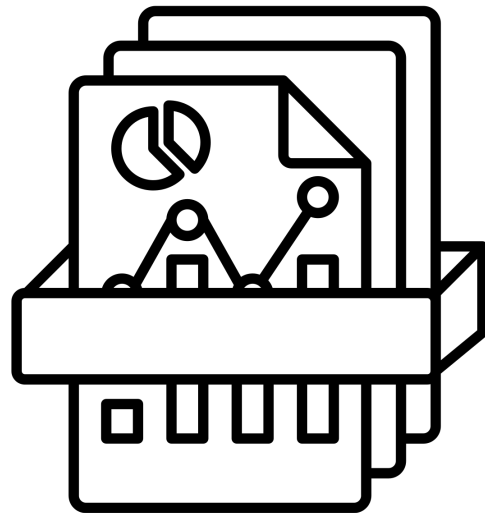
Key variables:

- Adjusted closing prices
- Trading volume
- Traded value

Adjusted prices incorporate:

- Dividends
- Stock splits
- Corporate actions

This ensures signals reflect true total return dynamics.



Data Analysis



Before constructing signals, we examine the statistical properties of the data.

Important observations:

- Equity prices often exhibit trend persistence
- Liquidity is concentrated in a small subset of stocks
- Liquid stocks support scalable systematic strategies

Common visualizations include:

- Price trend charts
- Moving average overlays
- Liquidity rankings



Features



Two core features are constructed from the price data.

Trend Indicators

- c_sma_50d (50-day Simple Moving Average)
- c_sma_200d (200-day Simple Moving Average)

These indicators smooth price movements and capture medium and long-term trends.

Liquidity Feature

- c_daily_traded_value_63d (3-Month Traded Value)

This metric ensures the strategy focuses on highly tradable securities.

Strategy Signals



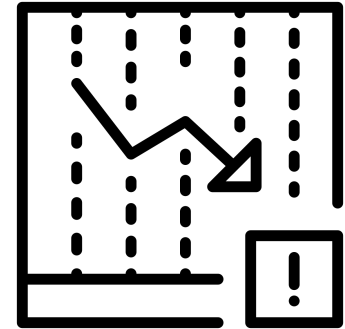
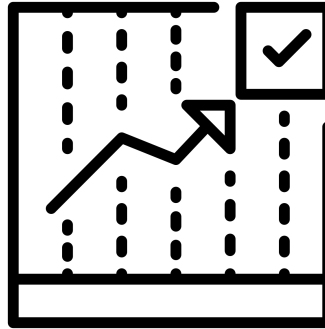
Trend signal definition:

Signal = 1 when

$SMA(50) > SMA(200)$

Signal = 0 when

$SMA(50) \leq SMA(200)$



Implementation in the Data Curator custom calculations = **c_sma_50d_200d_signal**

This rule identifies securities experiencing sustained upward momentum.

Signals are only generated once both moving averages have sufficient historical data.

Strategy Design

Liquidity-Weighted Trend Strategy

Strategy Modeling



Once signals are defined, we must determine how to allocate capital.

Portfolio construction requires answering three key questions.

Selection

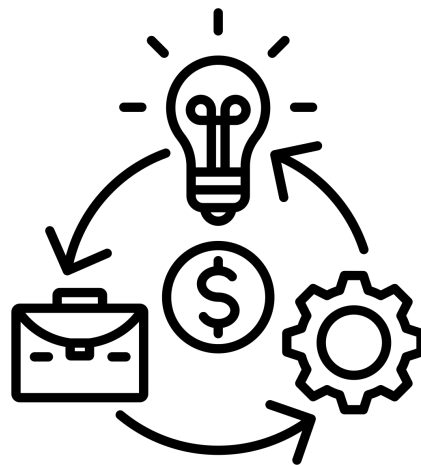
- Which stocks are trading above their 50-day and 200-day moving averages and rank among the Top 35 or X by 3-month traded value?

Sizing

- How should capital be allocated across the selected stocks based on their traded value, while respecting the 20% position limit?

Timing

- When does the composition of the Top X change, requiring the portfolio to rebalance?



Portfolio Construction



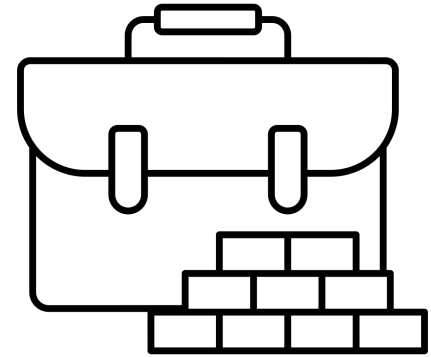
Selection rules:

- Securities with signal = 1
- Ranked by 3-month traded value
- Top securities selected

Sizing rules:

- Position weights proportional to traded value
- Maximum 20% weight per position

These rules define how capital is allocated across securities.



Portfolio Construction



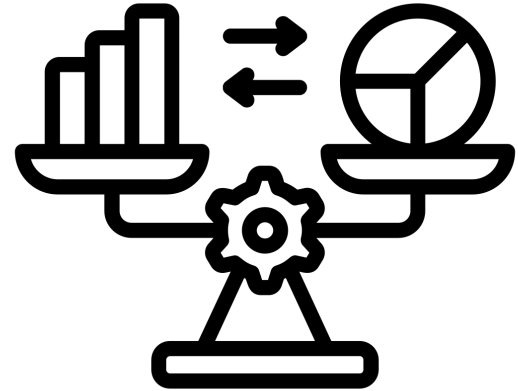
Rebalancing Logic

The strategy uses event-driven rebalancing.

Portfolio changes occur whenever:

- A security enters the Top X liquidity ranking
- A security exits the Top X liquidity ranking
- The number of eligible securities falls below X
- A portfolio constituent is delisted or becomes non-tradable

This ensures the portfolio continuously reflects the current signal and investable universe.



Portfolio Construction



Market Regime Handling

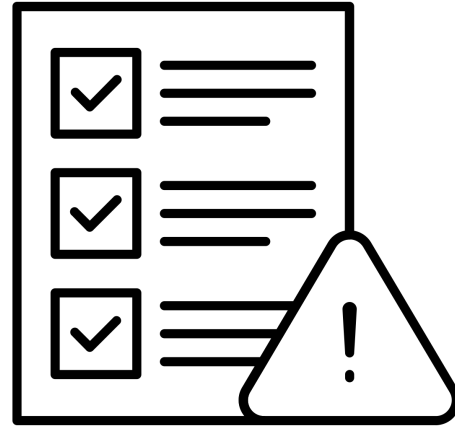
During weak markets, fewer stocks may satisfy the trend condition.

If the number of eligible stocks is less than 35, remaining capital is invested in:

SPY ETF - SPDR S&P 500 ETF Trust

This ensures:

- Capital remains invested
- Market exposure is maintained
- Diversification is preserved



Historical Portfolio Generation

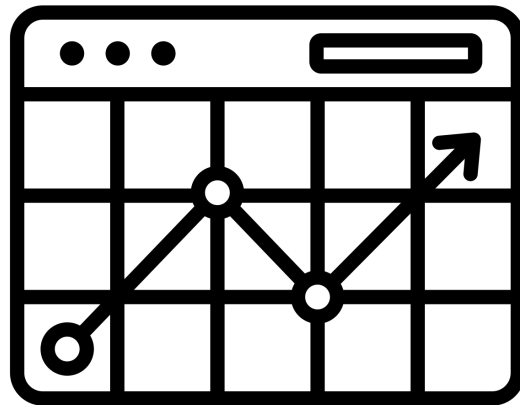


Applying the strategy rules through time produces a time series of historical portfolios.

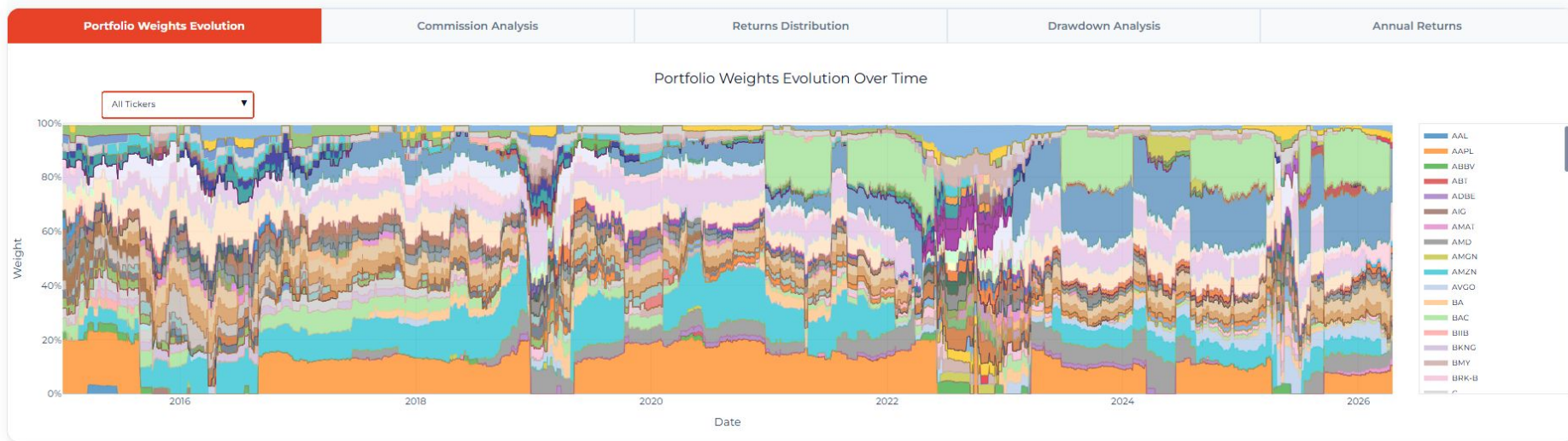
Each rebalance generates:

- Portfolio constituents
- Position weights
- Daily portfolio returns

These portfolios form the basis for performance evaluation.



Historical Portfolios Top 20



Backtesting

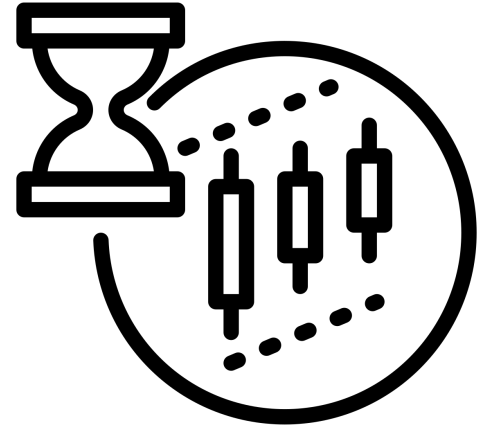


Backtesting evaluates how the strategy would have performed historically.

Key metrics include:

- Cumulative returns
- Volatility
- Drawdowns
- Turnover
- Risk-adjusted returns

However, **performance metrics alone are not sufficient to validate a strategy.**



Backtesting Config



parameter	value	description
portfolio_name	portfolio_weights	Name of the portfolio
benchmark_file_name	KN_US_Equity_Benchmark	Benchmark name
start_date	2015-01-01	From date, formatted as YYYY-MM-DD; e.g. 1999-12-31
end_date	2026-04-17	To date, formatted as YYYY-MM-DD; e.g. 2000-12-31
initial_capital	\$1,000,000.00	Initial capital to invest
cash_reserve_percentage	1.00%	Percentage of cash to reserve
commission_cents	\$0.05	Cents of every share to charge the transaction costs
input_market_data_directory	Data_Curator	Directory Path where your market data input files are located
input_portfolio_directory	Portfolio_Construction	Directory Path where your portfolio input files are located
backtest_results_output_directory	Backtest_Engine	Directory Path where your backtest result files are located
market_data_input_format	csv	Format of the prices files
portfolio_input_format	csv	Format of the portfolios files
commission_price_column	c_vwap	Column used to calculate the number of shares for commission fees (typically unadjusted VWAP)
trade_execution_price_column	m_vwap_split_adjusted	Column used as the trade execution price for buying/selling shares (typically split/dividend-adjusted VWAP)
mark_to_market_price_column	m_close_dividend_and_split_adjusted	Column used for daily portfolio valuation between rebalancing dates (typically split/dividend-adjusted close)
date_column	m_date	Column containing the trading dates
logger_level	info	Level of detail for the output logs; Shows all logs with severity equal or greater than the selected value
dashboard_port	8050	Port to expose for backtest dashboard

Performance Top 20



Backtest Dashboard

portfolio_weights

Analysis Period: 2015-01-02 to 2026-04-17

TOTAL RETURN

669.86%

ANNUALIZED RETURN

19.08%

SHARPE RATIO

0.976

SORTINO RATIO

1.293

ALPHA

5.515%

MAX DRAWDOWN

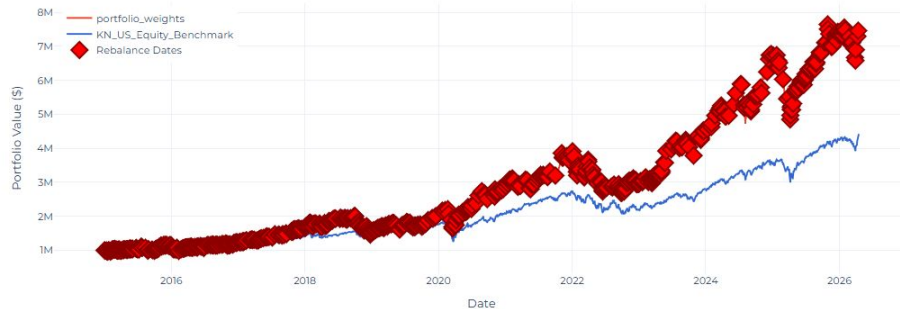
-32.36%

TOTAL COMMISSIONS

\$94,067

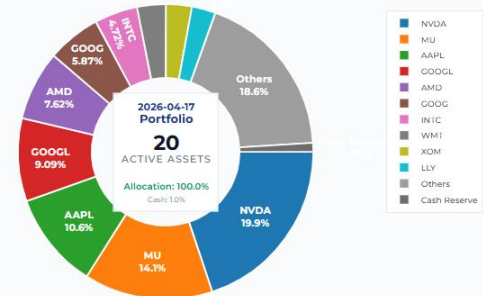
Portfolio Performance

Portfolio Performance Over Time



Portfolio Composition

Portfolio Composition - 2026-04-17 (Top 10)



Strategy Robustness

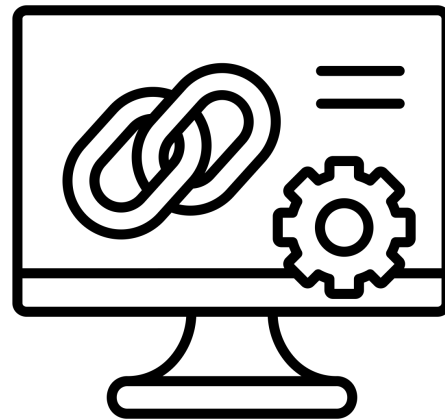


Robust strategies should be evaluated under different scenarios.

Key tests include:

- Out-of-sample performance
- Parameter sensitivity analysis
- Market regime evaluation
- Diversification analysis

These tests help identify strategies that are stable rather than overfit.





Drawdown Top 20



Annual Returns Top 20



Portfolio Weights Evolution				Commission Analysis		Returns Distribution		Drawdown Analysis		Annual Returns	
Annual Returns Comparison											
Portfolio vs Benchmark Annual Performance											
Year	year	start_date	end_date	portfolio_weights_return	portfolio_weights_start_value	portfolio_weights_end_value	KN_US_Equity_Benchmark_ret	KN_US_Equity_Benchmark_sta	KN_US_Equity_Benchmark_en	alpha	
0	2015	2015-01-02	2015-12-31	13.58%	\$999,070.45	\$1,138,786.87	2.35%	\$1,000,000.00	\$1,024,178.34	11.23%	
1	2016	2016-01-04	2016-12-30	9.76%	\$1,101,934.00	\$1,212,570.16	12.68%	\$1,008,196.21	\$1,139,771.56	-2.92%	
2	2017	2017-01-03	2017-12-29	33.84%	\$1,217,143.91	\$1,640,351.29	21.25%	\$1,149,892.74	\$1,400,709.16	12.58%	
3	2018	2018-01-02	2018-12-31	-6.40%	\$1,653,551.51	\$1,544,857.76	-4.70%	\$1,412,429.17	\$1,344,266.15	-1.70%	
4	2019	2019-01-02	2019-12-31	31.41%	\$1,537,638.67	\$2,036,056.73	30.01%	\$1,346,253.24	\$1,763,075.43	1.40%	
5	2020	2020-01-02	2020-12-31	38.16%	\$2,069,672.46	\$2,888,961.78	17.55%	\$1,779,288.26	\$2,102,226.99	20.62%	
6	2021	2021-01-04	2021-12-31	30.96%	\$2,885,232.11	\$3,806,970.41	30.11%	\$2,071,592.52	\$2,715,208.87	0.85%	
7	2022	2022-01-03	2022-12-30	-23.62%	\$3,910,959.26	\$2,964,730.79	-19.05%	\$2,733,695.10	\$2,199,926.69	-4.57%	
8	2023	2023-01-03	2023-12-29	51.49%	\$2,967,645.22	\$4,540,289.37	27.24%	\$2,192,472.97	\$2,805,823.18	24.24%	
9	2024	2024-01-02	2024-12-31	45.19%	\$4,455,659.07	\$6,546,090.45	25.48%	\$2,790,071.13	\$3,526,215.83	19.71%	
10	2025	2025-01-02	2025-12-31	13.23%	\$6,450,773.30	\$7,329,642.29	19.63%	\$3,518,945.15	\$4,230,749.15	-6.40%	
11	2026	2026-01-02	2026-04-17	15.91%	\$7,367,557.76	\$7,698,567.43	15.29%	\$4,239,874.62	\$4,423,286.18	0.62%	

Attribution Analysis

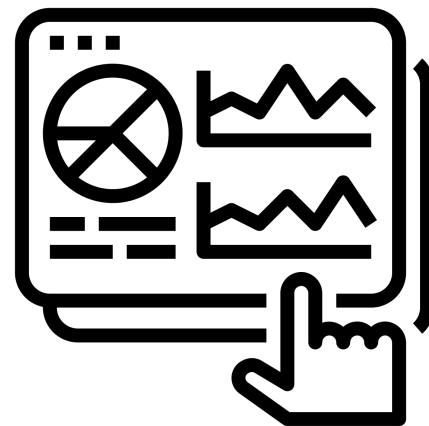


Performance must be decomposed to understand its true drivers.

Key attribution questions:

- How much performance comes from market exposure?
- How much from sector allocation?
- How much from stock selection?
- How much from factor exposures?

Attribution analysis **allows researchers to separate true alpha from unintended risk exposures.**



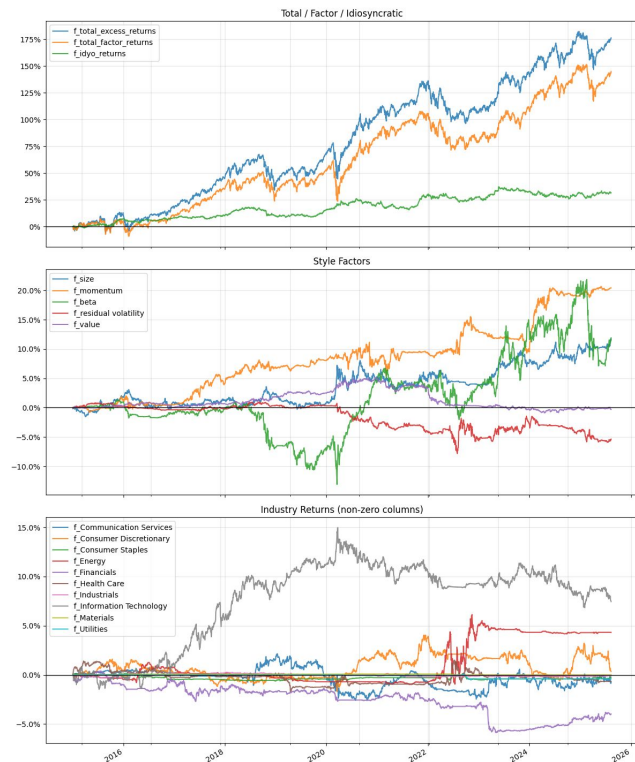
Attribution Analysis



1. Factor returns drive nearly all performance. Of the ~175% cumulative excess return, factor exposures account for ~150%, idiosyncratic (stock-picking) alpha contributes only ~25% and has flatlined since 2022.

2. Beta exposure is an unrewarded drag. The f_beta factor hit -12% during the COVID crash and still sits around -5% cumulatively. It adds drawdown risk without compensating return, neutralizing beta (constraining portfolio beta to ~1.0 vs. the benchmark) would immediately improve risk-adjusted performance.

3. Size and momentum are the real engines, but momentum needs guardrails. The size factor steadily contributes ~10–12%, while momentum adds ~5% with sharp reversals in 2020 and 2022. Adding a momentum-crash filter or regime-aware signal would preserve the upside while reducing those whipsaw drawdowns.



Scoring



1. We can improve the coverage of the investable universe using another data provider.
2. We need to be more clear to debug the logic to avoid rebalancing too often when we keep how many days need to pass to change the signal.
3. We can add more attribution analysis results to compare if the strategy actually improves with the ideas we discussed in that section.

Strategy Blueprint	Scoring Concept	Weight	Score (0-100)	Weighted Score	Gap (Score)
Strategy Definition	Hypothesis & Source of Returns	11.11%	100	11.1	0.0
	Research Foundation	11.11%	100	11.1	0.0
	Scientific Process	11.11%	100	11.1	0.0
Feature Engineering	Data Quality & Integrity	11.11%	66	7.3	-3.8
	Feature Design & Relevance	11.11%	100	11.1	0.0
	Signal Construction	11.11%	100	11.1	0.0
Strategy Design	Portfolio Construction Logic	11.11%	66	7.3	-3.8
	Backtesting & Robustness	11.11%	100	11.1	0.0
	Attribution & Understanding	11.11%	66	7.3	-3.8
Overall Score		100.00%		88.7	

Conclusions

Subtitle

Hypothesis Validation



Did the data support the source of returns?

Key Findings:

- The strategy outperforms the benchmark, confirming the presence of trend persistence
- Performance is primarily driven by factor exposures (momentum + size) rather than pure stock selection
- Returns are not constant across time — periods of underperformance align with known momentum crashes
- Beta exposure introduces unrewarded risk, reducing risk-adjusted performance

Hypothesis Validation



Interpretation:

- The hypothesis is partially validated
- The strategy captures real, known drivers of returns
- However, performance is not purely alpha — it is largely explained by systematic exposures

Conclusion:

The strategy works — but we now understand why it works, and where it breaks

Is This An Implementable Strategy?



Implementation Risks

Factor decay

- Momentum signals weaken as they become crowded

Market impact & liquidity

- Even with liquidity filters, execution costs can erode returns

Turnover & rebalancing

- Event-driven rebalancing may generate unnecessary trading

Capacity constraints

- Strategy scalability is limited to highly liquid securities

Is This An Implementable Strategy?



Operational Requirements

- Robust data pipelines and execution systems
- Continuous monitoring of signals and exposures
- Strict risk management framework

Decision

👉 Suitable for paper trading and controlled deployment

Key Condition

Implementation must include cost control, turnover constraints, and exposure management

Next Steps



1. Strategy Improvements

- Neutralize unrewarded beta exposure
- Introduce momentum crash protection / regime filters
- Enhance signals with additional features (multi-factor approach)

2. Implementation & Infrastructure

- Build production-grade data pipelines
- Implement execution-aware backtesting (costs, slippage)
- Deploy paper trading environment

Next Steps



3. Validation & Monitoring

- Track out-of-sample performance
- Monitor factor exposures and drift
- Define KPIs and alert systems

4. Scaling & Deployment

- Estimate capacity limits and liquidity impact
- Define capital allocation strategy
- Prepare for incremental live deployment

Contact

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